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REVIEW ARTICLE

Physical developer method for detection of latent fingerprints: A review

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Abstract The physical developer technique is a means to detect fingerprints on dry and wet, porous items, including paper articles, clay-based products and adhesive tapes. The process involves an oxidation–reduction couple whereby a solution of an iron salt reduces aqueous silver nitrate to finely divided metallic silver. The technique derives its name from the photographic physical developer which, during processing of film rolls, undergoes a similar redox reaction. The physical developer reveals the fingerprints as dark gray or black images due to the adsorption of metallic silver particles on the fatty acid and lipid components of sweat residue.
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1. Introduction

A latent fingerprint is formed when the sweat pores of the papillary ridges leave a deposition of perspiration on a surface with which the finger has been brought into contact.[1] Human body possesses the following three types of glands – *viz*. eccrine, apocrine and sebaceous, the secretions of which contribute to a fingerprint deposit.[2]

Eccrine glands are widely distributed throughout the body and are particularly numerous on the palms of hands and the soles of feet. These glands secrete a host of chemicals as fallout of metabolism and catabolism. Besides water content, eccrine sweat contains up to 1% of other substances of which inorganic salts constitute about one-half.[3] The other half consists of organic derivatives like, urea, creatinine, choline, lactic acid, sugars, uric acid, amino acids and proteins.[4]

The physical developer technique for detection of latent fingerprints is based on the interaction between lipids and fatty acid constituents of sweat residue, with colloidal silver particles.[5] It had earlier been known that colloidal silver particulates have affinity for such organic derivatives.[6] Since these biomolecules are essentially insoluble in water, the physical developer can detect latent impressions on wet porous items as well.[7] Moreover, the insoluble, organic ingredients of sweat are stable for prolonged periods of time, as a result of which fingerprints that are several years old may be visualized by this technique.[8,9] The developed prints are dark gray or black in color.[10]

2. Mechanism

The basis of the physical developer technique is the *in situ* reduction of silver(I) ions to colloidal, elemental silver which, in turn, interacts with the organic constituents of the fingerprint residue and ultimately visualizes the impressions as dark gray or black ridges.[11] The same redox reaction is involved in the development of photographic films.[12,13]

The physical developer composition is a mixture of the following components.[14]

(a) An aqueous solution of silver nitrate.
(b) An iron(II) and iron(III) redox couple.
(c) An acid buffer, usually citric acid.
(d) A cationic detergent, usually *n*-dodecylamine acetate.

The iron(II) ions reduce silver(I) ions to metallic silver, and simultaneously get oxidized to iron(III) ions:

$$Ag^{+} + Fe^{2+} \rightleftharpoons Ag^{0} + Fe^{3+}$$

Due to the reversible nature of this reaction, the chances are that iron(III) ions may re-oxidize metallic silver to silver(I) ions. To overcome this problem, citric acid is added to the fingerprint composition. The structure of citric acid is shown in Fig. 1.

Citric acid complexes with iron(III) ions and removes these from the equilibrium system, thereby pushing the redox reaction in forward direction. It also brings down the pH of the composition. This is important since optimum results are obtained under mildly acidic conditions.

The finely divided metallic silver particles, formed as a result of the redox reaction are colloidal in nature. It is in this state that these adhere to the fatty acid and lipid constituents of latent fingerprint and visualize the impression. However, if the concentration of metallic silver increases beyond a threshold, the particles coalesce and precipitate out. The colloidal nature of silver particles is lost and the quality of developed fingerprints tends to fall.

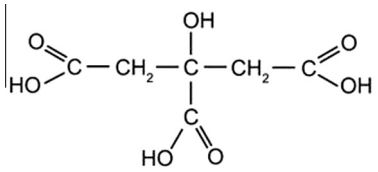


Figure 1 Structure of citric acid.

$$[CH_3(CH_2)_{10}CH_2NH_3]^{+}\ [CH_3COO]^{-}$$

Figure 2 Structure of *n*-dodecylamine acetate.

This problem is taken care of by incorporating a cationic detergent, usually *n*-dodecylamine acetate in the fingerprint composition. The structure of *n*-dodecylamine acetate is set out in Fig. 2.

The cations engulf silver particles, imparting a positive charge on to these. The positively charged particles repel each other, thereby averting the possibility of their coming close and condensing into a precipitate.[15] This phenomenon is called *peptization* and is symbolically shown in Fig. 3.

3. Pre-treatment of latent prints

The physical developer technique works very well on paper and paper products. However, the finished paper usually incorporates alkaline fillers and binders. These react with the physical developer composition and alter the concentration of its ingredients. It is therefore pertinent to neutralize the alkaline content of paper before attempting to develop fingerprints. This is carried out by giving an acid pre-wash to the paper sample.[16] Maleic acid, the structure of which is shown in Fig. 4, is commonly used for this purpose.

A solution containing 25 g maleic acid in 1 liter of water is prepared. The paper item is dipped in this solution for 5–10 min, till the bubbles cease to evolve. Thereafter, the procedure described in next section for development of fingerprints is followed.

Acetic acid[17] and malic acid[18] can serve as substitutes for maleic acids. Their structures are depicted in Fig. 4. However, chlorinated acids are unsuitable for neutralizing the alkali content of paper when physical developer technique of fingerprint detection is desired to be used. A pre-wash with dilute hydrochloric acid, for example, results in the formation of silver chloride which, almost instantaneously, decomposes into finely divided, black silver and darkens the background surface.[14]

4. Methodology

The following three solutions are required for detecting fingerprints by the physical developer method.[19]

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Figure 3 Peptization prevents colloidal silver particles from precipitating.

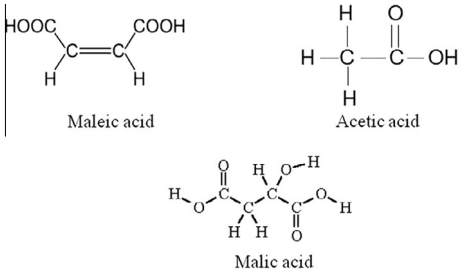


Figure 4 Pre-treatment reagents.

(i) *Solution A or the detergent solution*. It is prepared by dissolving 0.23 g *n*-dodecylamine acetate (Fig. 3) in 40 mL distilled water.
(ii) *Solution B or the redox solution*. It is prepared by adding successively, 30 g iron(III) nitrate, 80 g ammonium iron(II) sulfate (Mohr salt) and 20 g citric acid (Fig. 2) to 900 mL distilled water.
(iii) *Solution C or the silver nitrate solution*. It is prepared by dissolving 10 g silver nitrate in 50 mL distilled water. This solution is stored in a dark place.

The working solution is prepared as follows. Solution A is added to solution B. The mixture is stirred for about 2 min. Thereafter, solution C is added and stirring is continued for another 1–2 min. Before proceeding further, the authenticity of the working solution ought to be checked by the following test. A drop of dilute gold chloride solution is spotted on a Whatman filter paper. The latter is immersed in the working solution. If the spotted region gets darkened within 3–5 s, the solution is appropriate for use. Otherwise a fresh lot must be prepared.[20]

The working solution is next added to a glass tray. The tray is mounted on an electric shaker, one variety of which is shown in Fig. 5.

The object bearing the latent fingerprints is immersed in the working solution which is then gently agitated by the shaker



Figure 5 An electric shaker.

for about 15 min. The tray should not be overloaded with samples. Preferably, only one item should be processed at a time.[21] Dark gray or black colored fingerprints develop.

The object is thoroughly washed by immersing it in distilled water for 5 min. The washing operation is repeated twice more, each time with a fresh batch of distilled water.

5. Post-treatment of developed prints

The fingerprints processed by physical developer composition may be enhanced by post-treating these with a 2–5% solution of sodium hypochlorite, NaOCl. The silver particles adsorbed on the fatty acid and lipid constituents of latent impressions are oxidized to silver(I) oxide, Ag_2O by the hypochlorite ions:

$$OCl^- + 2Ag \rightarrow Ag_2O + Cl^-$$

The enhancement of developed fingerprints comes about by a twofold action. Firstly, silver(I) oxide is darker in shade as compared to finely divided metallic silver. Hence the ridge detail appears more sharp and pronounced. Secondly, sodium hypochlorite has bleaching properties, as a result of which it lightens the background color of the surface in question, markedly increasing the contrast.[8] This process is called *bleach toning*.

Yet another post-treatment procedure, called *potassium iodide toning* may be followed when fingerprints are to be developed on dark or patterned surfaces. In this method, the redox solution is mixed with a 20% w/v aqueous solution of potassium iodide. The iron(III) content of the redox solution reacts with iodide ions to yield iodine:

$$2Fe^{3+} + 6I^- \rightarrow 2FeI_2 + I_2$$

The combination of iodine with iodide ions furnished by potassium iodide produces triiodide ions:

$$I_2 + I^- \rightarrow I_3^-$$

The triiodide ions react with the starch content of paper items and produce dark blue background coloration. The silver adsorbed on the ridges combines with residual iodide ions to yield yellow colored silver iodide. The yellow fingerprint pattern on a dark blue background is easily discernible.[5]

Another post-treatment procedure involves conversion of adsorbed silver to silver sulfide by its reaction with radioactive sodium sulfide. Since the ridges become lined up with radioactive silver sulfide, the pattern may be imaged by autoradiography technique.[22] However, dealing with radioactive materials requires a great deal of precautions. A simpler procedure is to convert silver into a fluorescent complex, illuminate the impression with radiation having a wavelength around 515 nm and photograph it.[23]

6. Assessment of physical developer method

The physical developer composition detects latent fingerprints on the following items: Latex and rubber gloves; sticky and non-sticky sides of adhesive tapes; raw wood; unglazed

porcelain; rayon and nylon clothing; and different varieties of papers, including newspapers, photostat papers and ruled papers. It can also develop shoeprint impressions.[24]

The composition tags the fatty acid and lipid components of sweat residue for visualizing fingerprints. Since these biomolecules are insoluble in water, the physical developer technique lifts fingerprints impinged on moist items as well. It also develops fingerprints on surfaces which have been exposed to intense sunlight for several hours.

The technique presents some problems as well. It is time consuming and expensive, although a cost-effective version is also available.[25] The water used to prepare the working solution should be ultrapure. In fact, for proper results, it is purified by reverse osmosis, followed by de-ionization.[26] The working solution has a relatively short shelf life. The fingerprint processing procedure is destructive and leaves permanent stains on documents, especially if the latter are held with tongs/forceps. Clips and pins too must be removed from the documents before processing these for fingerprint detection.

7. Conclusion

Physical developer is a sensitive method of detecting fingerprints on porous items. It is specifically useful when fingerprints are required to be visualized on articles which have been deliberately or accidentally wetted. Under these conditions, the amino acid constituents of sweat residue are easily washed out and the ninhydrin method fails. Nevertheless, physical developer yields results. The sensitivity of the composition may be enhanced by pre-treatment and/or post-treatment methodologies. However, the requirement of a relatively large number of reagents makes the technique cumbersome and expensive.

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